

BSESS-F26-1A

MA-216

Discrete Structure

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AIR UNIVERSITY

Assignment. 01

Truth table for each Law.

Week: 02 Lecture: 05, 06

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Assign: 24 sep 26, thu

Done: 29 sep 26, tue

Due: 02 oct 26, thu

p	q	r	$p \wedge q$	$q \wedge r$	$p \vee q$	$q \vee r$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	T	T	F
T	F	T	F	F	T	T	F
T	F	F	F	F	T	F	F
F	T	T	F	T	T	T	F
F	T	F	F	F	T	T	F
F	F	T	F	F	F	T	F
F	F	F	F	F	F	F	F

$$(p \vee q) \vee r \equiv p \vee (q \vee r)$$

T	T
T	T
T	T
T	T
T	T
T	T
T	T
F	F

*₃. Distributive Laws:

$$\rightarrow \text{(~~p \wedge q~~) } p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$$

- I will enter Uni AND (I have my ID OR live auth pass.)
- (I enter AND have ID) OR (I enter AND have auth pass)

$$\rightarrow p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$$

- Firewall blocks connection OR (ip is blacklisted AND connection is untrusted)
- (firewall blocks conn. OR ip is black) AND (firewall blocks conn. OR conn. is untrusted)

	$q \vee r$	$p \wedge q$	$p \wedge r$	$p \wedge (q \vee r)$	$(p \wedge q) \vee (p \wedge r)$	$q \wedge r$	$p \vee q$	$p \vee r$	$p \vee (q \wedge r)$	$(p \vee q) \wedge (p \vee r)$
1.	T	T	T	T	T	T	T	T	T	T
2.	T	T	F	T	T	F	T	T	T	T
3.	T	F	T	T	T	F	T	T	T	T
4.	F	F	F	F	F	F	T	T	T	T
5.	T	F	F	F	F	T	T	T	T	T
6.	T	F	F	F	F	F	T	F	F	F
7.	T	F	F	F	F	F	F	T	F	F
8.	F	F	F	F	F	F	F	F	F	F

$\uparrow \equiv \uparrow$ $\uparrow \equiv \uparrow$

*4. Identity Law:

$$\rightarrow p \wedge t \equiv p$$

• It's raining AND $1+1=2$

$$\rightarrow p \vee c \equiv p$$

• It's raining OR $1=0$.

p	t	$p \wedge t \equiv p$
T	T	T
F	T	F

p	c	$p \vee c \equiv p$
T	F	T
F	F	F

*5. Negation law:

$$\rightarrow p \vee \sim p \equiv t$$

$2+2=4$ OR $2+2 \neq 4$

$$\rightarrow p \wedge \sim p \equiv c$$

It is true OR it is not true.

p	$\sim p$	$p \vee \sim p \equiv t$
T	F	T
F	T	T

p	$\sim p$	$p \wedge \sim p \equiv c$
T	F	F
F	T	F

* Double Negation law:

→ $\sim(\sim p) \equiv p$

It's true : p

It's not true : $\sim p$

It's NOT not true : $\sim(\sim p)$

p	$\sim p$	$\sim(\sim p) \equiv p$
T	F	T
F	T	F

* Idempotent Law:

→ $p \wedge p \equiv p$

It's true AND It's true.

→ $p \vee p \equiv p$

It's true OR It's true.

* De Morgan's Law:

→ $\sim(p \wedge q) \equiv \sim p \vee \sim q$

p: User has valid passcode.

q: User has MFA.

$\sim(p \wedge q)$: It is not the case that (User has valid pass AND MFA).

$\sim p \vee \sim q$: User NOT have pass OR mfa is NOT enabled.

p	q	$\sim p$	$\sim q$	$p \wedge q$	$\sim(p \wedge q)$	$\sim p \vee \sim q$
T	T	F	F	T	F	F
T	F	F	T	F	T	T
F	T	T	F	F	T	T
F	F	T	T	F	T	T

→ $\sim(p \vee q)$

p: Firewall

$\sim(p \vee q)$:

$\sim p \wedge \sim q$:

p	q
T	T
T	F
F	T
F	F

* Universal

→ $p \vee t$

• I g

→ $p \wedge c$

• Serv

* Abs

→ $p \vee$

p
T
T
F
F

$$\sim(p \vee q) \equiv \sim p \wedge \sim q$$

p : Firewall is enabled. q : IDS is running.

$\sim(p \vee q)$: It is NOT true that (firewall is enabled OR ids is running).

$\sim p \wedge \sim q$: Firewall is NOT enabled AND IDS is ~~NOT~~ NOT running.

p	q	$\sim p$	$\sim q$	$(p \vee q)$	$\sim(p \vee q)$	$\sim p \wedge \sim q$
T	T	F	F	T	F	F
T	F	F	T	T	F	F
F	T	T	F	T	F	F
F	F	T	T	F	T	T

$\begin{matrix} \text{---} & \text{---} \\ | & | \\ \text{---} & \text{---} \end{matrix} \equiv \begin{matrix} \text{---} & \text{---} \\ | & | \\ \text{---} & \text{---} \end{matrix}$

* 9. Universal Bound law:

$$\rightarrow p \vee t \equiv t$$

p	t	$p \vee t \equiv t$
T	T	T
F	T	T

• I got 10/10 in this assignment OR Allah is one.

$$\rightarrow p \wedge c \equiv c$$

p	c	$p \wedge c \equiv c$
T	F	F
F	F	F

• Server is compromised AND 1=0.

* 10. Absorption law:

$$\rightarrow p \vee (p \wedge q) \equiv p$$

p	q	$p \wedge q$	$p \vee (p \wedge q)$
T	T	T	T
T	F	F	T
F	T	F	F
F	F	F	F

• User is authenticated OR (User is authenticated AND has MFA enabled.)

$$\rightarrow p \wedge (p \vee q) \equiv p$$

p	q	$p \vee q$	$p \wedge (p \vee q)$
T	T	T	T
T	F	T	T
F	T	T	F
F	F	F	F

• firewall is enabled AND (firewall is enabled OR IDS is enabled.)

11. Negation of t and c:

$$\neg t \equiv c$$

t	$\neg t$	c
T	F	F

t: I am alive

c: I am not alive

$\neg t$: I am not alive.

$$\neg c \equiv t$$

c	$\neg c$	t
F	T	T

c: This is NOT true.

t: This is true.

$\neg c$: This is true.

———— The End ————